

CAMPINA, Romania

WMO: 153490

Lat: 45.1436N

Lon: 25.7333E

Elev: 462

StdP: 95.89

Time Zone: 2.00 (EUE)

Period: 87-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-9.4	-15.6	1.0	-9.1	-13.6	1.2	-7.0	12.0	0.5	7.3	1.5	0.9	0	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	30.3	20.0	29.0	19.5	27.6	19.1	21.4	28.1	20.6	27.1	19.8	26.0	2.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.8	24.1	18.4	14.0	23.3	17.6	13.3	22.6	64.5	28.6	61.4	27.3	58.7	26.2	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	5.4	4.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-0.9	1.5	4.8	9.5	14.8	18.9	20.1	20.2	15.6	9.9	4.9	1.4
	DBStd	8.35	4.56	4.50	4.38	4.28	3.10	2.80	2.57	2.84	3.66	3.89	4.38	4.31
	HDD10.0	1287	337	239	167	61	3	0	0	0	2	48	162	267
	HDD18.3	3205	596	471	419	266	114	26	13	16	96	261	404	524
	CDD10.0	1327	0	1	6	46	152	268	313	316	170	46	8	1
	CDD18.3	204	0	0	0	0	5	44	67	73	14	0	0	0
	CDH23.3	1987	0	0	0	16	83	419	588	701	173	8	0	0
	CDH26.7	510	0	0	0	1	13	88	152	219	37	0	0	0
Wind	WSAvg	1.5	1.3	1.5	1.9	1.9	1.7	1.6	1.6	1.4	1.5	1.4	1.4	1.3
Precipitation	PrecAvg	795	40	34	42	65	93	112	101	80	68	62	48	50
	PrecMax	1108	87	86	79	168	199	186	220	244	138	138	105	133
	PrecMin	459	9	6	7	28	21	46	40	21	22	4	2	4
	PrecStd	159	21	19	21	34	44	38	46	46	33	32	27	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	17.1	21.0	25.2	28.3	30.8	32.1	32.3	29.7	24.5	19.4	16.4
		MCWB	7.7	10.1	12.3	14.5	17.8	20.9	21.7	20.5	17.8	15.3	12.4	9.4
	2%	DB	10.2	13.4	17.9	22.6	25.5	28.9	29.9	30.5	27.4	22.2	16.4	12.4
		MCWB	5.7	7.4	9.9	13.8	16.5	19.8	20.6	19.7	17.7	14.8	10.9	7.3
	5%	DB	7.5	10.8	15.2	20.3	23.9	27.4	28.4	29.1	25.1	19.7	13.7	9.7
		MCWB	4.0	6.4	8.8	12.8	15.9	19.5	19.9	19.3	16.9	13.2	10.0	6.3
	10%	DB	5.1	8.0	12.5	17.8	21.8	25.8	26.9	27.3	23.1	17.0	11.1	7.3
		MCWB	2.7	4.6	7.3	11.0	15.0	18.7	19.4	19.1	16.1	11.8	8.6	4.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	10.5	12.5	15.5	18.6	22.5	22.5	22.5	20.0	16.6	13.2	9.6
		MCDB	13.3	16.8	19.7	23.2	25.4	28.8	29.4	30.0	27.8	22.0	18.1	13.5
	2%	WB	6.0	8.2	10.5	14.3	17.5	21.1	21.5	21.1	18.5	15.0	11.8	8.3
		MCDB	9.4	12.5	16.6	22.1	23.8	27.6	28.1	28.1	25.0	20.6	15.3	11.9
	5%	WB	4.3	6.8	9.1	13.1	16.6	20.1	20.7	20.2	17.5	13.8	10.4	6.6
		MCDB	6.8	10.4	14.1	19.2	22.1	26.2	26.9	26.9	23.6	19.3	12.9	9.3
	10%	WB	2.9	5.3	7.7	11.9	15.5	19.2	19.8	19.4	16.6	12.6	9.1	5.1
		MCDB	4.6	7.8	11.5	16.8	20.6	25.1	25.6	25.8	22.4	16.2	11.0	7.6
Mean Daily Temperature Range	5% DB	MDBR	8.4	9.4	10.1	10.9	10.7	11.3	11.4	12.0	11.4	10.5	8.4	8.4
		MCDBR	12.0	13.9	15.2	15.5	14.2	13.5	13.7	14.7	14.8	15.4	12.3	12.4
		MCWBR	7.5	8.8	8.3	7.8	6.7	6.4	6.1	5.8	6.3	8.1	7.3	7.5
	5% WB	MCDBR	11.1	13.2	13.9	14.3	12.4	12.6	12.3	12.9	13.3	14.5	10.6	11.3
		MCWBR	7.2	8.3	8.1	7.4	6.5	6.3	6.1	5.7	5.9	8.1	6.6	7.1
Clear-Sky Solar Irradiance	taub		0.320	0.352	0.394	0.456	0.432	0.435	0.448	0.466	0.419	0.380	0.352	0.322
	taud		2.313	2.223	2.171	2.033	2.175	2.206	2.178	2.118	2.214	2.305	2.336	2.336
	Ebn at Noon		776	813	827	804	840	837	820	787	793	773	730	731
	Edh at Noon		80	106	128	161	144	140	143	146	122	97	79	72
All-Sky Solar Radiation	RadAvg		1.38	2.20	3.34	4.46	5.49	5.93	6.02	5.41	3.95	2.65	1.50	1.13
	RadStd		0.17	0.30	0.35	0.47	0.48	0.48	0.46	0.39	0.35	0.33	0.17	0.15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (70 neighbors)	+0.53	N/A	N/A	+0.65	N/A	N/A	N/A	-153	+76	+36

Nomenclature: See separate page